

# NERD 2026 Application

**Status:** Updated to match restructured WP plan (WP\_detailed\_plans.md, “New plan” section). Written for Committee for Natural and Technical Sciences audience. **Deadline:** 19 February 2026, 14:00 CET

## Timeline

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**NB: gantt\_chart.yaml is the authoritative timeline and should be matched by the budget**

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## Potential Figures

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1. **Figure 1:** Conceptual drive model — the cyclical arms race: X drive ↔ autosomal suppression ↔ Y drive, each step with collateral neurodevelopmental effects (DriveModel 2.pdf)
2. **Figure 2:** Multi-panel figure: (a) Three-way Venn diagram with gene lists in each overlap group, gene names colored if at A/B borders; (b) Expression trajectories for HUWE1 and RBMX2
3. **Figure 3:** A/B eigenvector plot showing X chromosome compartments in sperm with RBMX2 at the A/B border, including zoom panel (ab\_compartments\_with\_rbm2.pdf)
4. **Figure 4:** Project timeline (Gantt chart) with vertical lines connecting work packages at expected result-exchange points

## 1. PROJECT TITLE (max 150 characters)

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Mind the Drive: The Neurodevelopmental Cost of Policing Sex Chromosomes

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## 2. BRIEF PROJECT DESCRIPTION (max 2,000 characters)

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The sex chromosomes harbor an evolutionary conflict invisible to standard genetic studies. When a gene on the X chromosome escapes silencing in sperm cells, it can bias its own transmission to the next generation — a process called meiotic drive. Because an even sex ratio is a game-

theoretic equilibrium, such a driver immediately triggers counter-evolution: variants on the Y and autosomes that restore balanced transmission are promoted by natural selection even when they are mildly deleterious. The result is a self-erasing arms race that leaves no sex-ratio distortion but continuously elevates the population frequency of harmful alleles — including alleles affecting brain development.

This project proposes that this conflict is a major unrecognized force shaping the genetics of autism spectrum disorder (ASD), the condition with one of the strongest sex biases in human genetics (4:1 male-to-female). The hypothesis makes testable predictions, and pilot data already support several: genes that are both differentially expressed between X- and Y-bearing sperm, under recent positive selection, and associated with ASD overlap far more than expected by chance. A key weapon in this arms race is the Y chromosome's bulk of repetitive DNA: human Y chromosomes vary dramatically in heterochromatin content between paternal lineages, and this structural variation can sequester chromatin-modifying proteins, altering which genes elsewhere in the genome are active — a "sink" effect demonstrated in model organisms and now linked, in our preliminary analyses, to lineage-dependent differences in ASD risk.

The project combines Markov jump models and phase-type distribution theory for molecular evolution, Bayesian regression for genetic association, probabilistic programming and Eigen analysis for 3D chromatin inference, with deep convolutional neural networks for polygenic risk score prediction - all simultaneous efforts required to address the proposed hypothesis.

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## 3. PROJECT DESCRIPTION (max 30,000 characters)

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### Background 1

**Meiotic drive.** During the production of sperm cells (spermatogenesis), a precursor cell containing both X and Y divides to produce two types of haploid cells: sperm carrying the X, and sperm carrying the Y. Unlike all other chromosomes, X and Y sex chromosomes only recombine at their ends, allowing them to evolve independently. The lack of genetic recombination allows genes on the X and Y chromosomes to affect their own transmission to the next generation. A gene on the X or Y chromosome may distort the even transmission of X and Y carrying germ cells if it is able to influence the survival or functioning of sperm cells

carrying the other sex chromosome. This phenomenon is called meiotic drive. It follows the mathematical logic of natural selection, but allows for strong selection on “selfish” driver genes on the sex chromosomes even if lowering organism fitness – e.g., reducing fertility or compromising neurodevelopmental integrity. That meiotic drive is a general mechanism shaping genome evolution follows from the empirically supported mathematical models of molecular evolution, in much the same way that the Standard Model in physics predicted the Higgs boson before its existence was confirmed. However, even though meiotic drive has been observed in fungi, insects, and mammals ([Lindholm et al. 2016](#)), biologists seem reluctant to apply such mathematical logic to human evolution in the absence of empirical evidence. With this project, I formulate two hypotheses that link meiotic drive to the genetic basis of Autism Spectrum Disorder (ASD). By linking the evolutionary model to an observable phenotype (ASD) it becomes possible to formulate testable predictions that can be matched by empirical observations.

**The link to ASD.** This project proposes that the intragenomic conflict between selfish driver genes and the remaining genome is a major but unrecognized force shaping the genetic architecture of autism spectrum disorder (ASD). It is the condition with one of the strongest sex biases in human genetics (4:1 male-to-female ratio, affecting 3% of Danish children). The prevailing model assumes that the population frequencies of ASD-associated variants are controlled only by negative natural selection. I propose instead that selfish sex-linked driver genes and the compensatory evolution they provoke across the genome continuously generate and elevate ASD-relevant variation as collateral damage of an ongoing evolutionary arms race — a fundamentally non-adaptive process with direct consequences for neurodevelopment. Proteomic surveys show that roughly 80% of brain proteins are also present in testis, and approximately one in four genes active in neurons is also expressed in spermatids ([Matos et al. 2021](#)). This overlap creates an evolutionary conflict: a gene variant may be selected for its advantage in spermatogenesis despite being mildly deleterious for brain development. Meiotic drive turbo-charges this conflict: The natural selection on an X chromosome gene conferring a 10% transmission advantage of X-carrying sperm is on par with the strongest natural selection observed in humans (lactase persistence). In addition to the ASD risk associated with the driver itself, any ASD risk variants in the genomic neighborhood will “hitchhike” to higher population frequency along with the driver.

**Why drive is hard to observe directly.** A natural objection is that transmission distortion and sex-ratio bias have not been reported in humans. This is expected, because an even sex ratio is a game-theoretic

equilibrium — Fisher's principle ([Fisher 1930](#)): when one sex is rarer, individuals producing more offspring of that sex gain more grandchildren, so any variant restoring even transmission is strongly favored by natural selection, even if that variant is mildly deleterious in other respects. A driver distorting X/Y transmission therefore triggers immediate counter-selection: existing variants on the Y or the non-sex chromosomes that restore balanced transmission are swept to high frequency, and new mutations with the same effect are favored as they arise (Figure 1). Drive is therefore transient: it flares, pushes the driver (and its hitchhiking neighborhood) to high frequency, and then subsides as suppressors equilibrate the sex ratio — leaving no ongoing bias but a lasting footprint of elevated allele frequencies. Crucially, suppressors are themselves under strong positive selection and drag along their own hitchhiking neighborhoods of mildly deleterious alleles — the arms race generates collateral damage on both sides. This transient, self-erasing dynamic means drive is best detected through its evolutionary signatures (selective sweeps, hitchhiking) rather than through contemporary transmission ratios.

## Hypothesis 1: Meiotic Drive on the X Chromosome

*Meiotic drive on the sex chromosomes and the counter-evolution it triggers on the Y and autosomes continuously generate and elevate ASD-relevant genetic variation as collateral damage of an arms race for transmission advantage in sperm.*

**Dual-function neuron–sperm genes carry excess ASD risk** A pilot gene-set analysis of a large Danish clinical cohort shows that neuron genes co-expressed in spermatids are significantly enriched for ASD association in males ( $p = 0.030$ ), while neuron genes without spermatid expression show no enrichment — confirming that specifically the dual-function genes carry risk.

**Among brain genes, selection targets those co-expressed in spermatids** A pilot analysis of evolutionary trees (genealogies) reconstructed from population genomic data identifies 35 X-linked genes under recent positive selection; among these nine known ASD genes, representing a highly significant enrichment (fisher  $p$ -value  $5.95e-5$ ). We further find that among brain-expressed genes, selection is significantly more frequent among genes also expressed in spermatids ( $p = 0.002$ ) supporting the prediction that selection is associated with spermatid expression rather than neuron expression.

**ASD genes are differentially expressed between X- and Y-carrying sperm** Post-meiotic spermatids are connected by intercellular bridges mediating selective directional transport ensuring that RNA gene transcripts are equally distributed. Only gene variants able to disable this redistribution can act as drivers. Using available data on single-cell RNA expression through spermatogenesis in a pilot study, we find that known autism genes (SFARI database) are significantly enriched among genes differentially expressed between X-bearing and Y-bearing spermatids ( $p = 0.004$ ).

**Candidate meiotic drivers are ASD genes** Genes differentially expressed between X- and Y-carrying spermatids (potential drivers) also under recent positive selection (like active drivers would be) significantly overlap with ASD genes (SFARI database). The pairwise overlaps between genes under recent positive selection, differentially expressed genes, and ASD genes are each statistically significant. This is also true for three-way overlap (Figure 2) that yield five genes: CDKL5, CLCN4, HUWE1, IL1RAPL1, PTCHD1. Among these candidate driver genes, HUWE1 is the clearest embodiment of the neuron–spermatid conflict. In the testis, HUWE1 is essential for chromatin remodeling during spermiogenesis — knockout causes meiotic arrest and complete infertility. In the brain, the same gene regulates neural progenitor differentiation; loss-of-function mutations cause intellectual disability with autistic features.

## Background 2

**A strong paternal effect.** A half-sibling study in 407 male rhesus macaques found that social functioning — scored on a reverse-translated human autism trait scale — is strongly heritable through fathers but not through mothers ([Garner and Capitanio 2023](#)). Because sons inherit their Y chromosome exclusively from their father, a global chromatin-modulating role of Y heterochromatin would produce exactly this pattern: paternal genotype shapes the chromatin environment in which autosomal and X-linked risk variants operate, while maternal genotype does not. The macaque finding therefore provides independent, cross-species support for the hypothesis that Y chromosome structure is a major modifier of ASD-related traits.

**The heterochromatin sink.** The second hypothesis concerns the Y chromosome not as a carrier of protein-coding genes — it has very few — but as a structural element that sequesters chromatin-modifying proteins. Large portions of the Y consist of repetitive DNA packaged as heterochromatin, a condensed form of chromatin associated with gene silencing. Maintaining heterochromatin requires specific proteins that exist

in limited quantities in the nucleus. A Y chromosome carrying more heterochromatin will sequester more of these proteins, leaving fewer available to maintain heterochromatin elsewhere in the genome. This “sink” effect has been rigorously demonstrated in *Drosophila*, where adding or removing Y chromosomes causes measurable genome-wide redistribution of histone marks at the boundaries between heterochromatin and euchromatin (the active form of chromatin) ([Brown, Nguyen, and Bachtrog 2020](#)).

**Variation between human Y haplogroups.** Recent telomere-to-telomere assemblies of 43 human Y chromosomes ([Rhie et al. 2023](#)) reveal striking structural variation between Y lineages (haplogroups). Centromeric repeat arrays vary more than 2-fold in size between major haplogroups (e.g. haplogroup R1b: mean 341 kbp; haplogroup I: mean 787 kbp). The DYZ1 satellite repeat varies over an order of magnitude (7–98 Mb). In Denmark, most men carry a Y from either haplogroup I or haplogroup R, making the Danish clinical cohort an ideal testing ground for haplogroup-dependent effects. If the sink model operates in humans, the hypothesis predicts that haplogroup I males (more Y heterochromatin) should show weakened heterochromatin boundaries elsewhere in the genome — potentially exposing genetic variants in normally silenced regions near centromeres. Results of genome-wide association studies (GWAS) should differ systematically between haplogroup backgrounds, not because the autosomal variants themselves differ, but because the chromatin context determining their activity differs. This is a form of epistasis (gene-gene interaction) that standard GWAS, which ignores the Y, cannot detect.

**Chromatin compartment analysis.** Hi-C (high-throughput chromosome conformation capture) measures the frequency of physical contact between all pairs of genomic loci, producing a genome-wide contact count matrix that reflects the three-dimensional folding of chromosomes in the nucleus. The first eigenvector of the normalized contact covariance matrix partitions the genome into two compartments: A compartments (positive eigenvector values) correspond to open, gene-rich, transcriptionally active chromatin, while B compartments (negative values) correspond to condensed, gene-poor, silenced chromatin. Genes located at the boundaries between A and B compartments are the most vulnerable to the sink effect: a small shift in the availability of chromatin-modifying proteins — caused by a Y chromosome that sequesters more or fewer of them — can tip boundary genes from active to silent or vice versa.

## Hypothesis 2: The Y Chromosome as a Chromatin



## Regulator

*Y chromosome repeat content evolves as the Y's weapon in the meiotic drive arms race: by sequestering limited chromatin-modifying proteins, a Y chromosome can shift which genes are accessible in sperm — and, as collateral damage, across the rest of the genome, creating paternal-lineage-dependent differences in ASD susceptibility.*

**ASD association depends on Y-haplotype.** A pilot study stratifying males in the Danish clinical ASD cohort by Y haplogroup reveals that different variants reach statistical significance depending on haplogroup background: Such context-dependent association matches the prediction of the sink model hypothesis: The same variant has different penetrance depending on whether the Y chromosome creates a strong or weak heterochromatin sink. Our pilot study covers only the X chromosome but finds that the RBMX2 gene shows significant association with ASD among males carrying Y chromosomes from haplogroup I, but not among haplogroup R males — despite the R cohort being larger, ruling out a simple power explanation. Crucially, RBMX2 sits precisely at the border of a chromatin compartment, exactly where the sink model predicts Y heterochromatin content will modulate gene accessibility (Figure 3). If haplogroup I's greater heterochromatin load shifts this boundary, RBMX2 could move from active to silenced chromatin, altering its expression and its contribution to ASD risk.

**RBMX2 bridges hypothesis 1 and 2.** It turns out RBMX2 may also be a poster child for Hypothesis 1 as depicted in Figure 1: It not only has a paralog on the Y chromosome (RBMX2) but also a retro-transposed copy (RBMXL2) on the chromosome 11. These genes encode splicing-fidelity factors, thus providing each player in the three-way X-Y-Autosome arms a means to influence broad downstream effects on transcript integrity in the germline.

## Research Plan

The project is organized around four work packages, each led by one person with the core expertise required. However, because the work packages require extensive exchange of results, the overlap of personnel is designed to ensure the right combination of skills available at the appropriate time. In the time line (Figure 4), arrows connecting work packages represent the most important exchanges of results. Where possible, postdocs and two PhD students have staggered start dates to spread the recruiting effort and startup supervision.

### Work Package 1: Y-stratified genome-wide association analysis

**incl. the X chromosome.** This package performs the first systematic inclusion of both sex chromosomes in ASD genetic association studies. We begin with sex-stratified GWAS and X chromosome-specific association analysis in the Danish clinical cohort with more than 17,000 male and 5,500 female cases. Data access is provided through an established and ongoing collaboration with a psychiatric genetics group that maintains the genotyping infrastructure and data governance. Standard GWAS software assumes diploid autosomes; the X requires separate treatment of males (hemizygous) and females, where one copy of most genes is silenced. Sex-stratified analyses of more than 17,000 male ASD cases will quantify X-linked heritability and test whether this is concentrated in the gene sets predicted by the drive hypothesis: spermatid/neuron overlap genes, differentially expressed genes, and genes hitchhiking to higher frequency in the genomic neighborhoods of past drivers.

We will stratify the male cohort by the major Y haplogroups I and R (identified from diagnostic SNPs on the genotyping panel) and perform independent autosomal GWAS within each stratum. We will compute the genetic correlation for ASD between haplogroup I and haplogroup R males: a low correlation would be direct evidence that the two Y backgrounds create distinct genetic architectures for the same disorder. We will quantify missing heritability accounted for by epistatic effects of the Y chromosome using a Bayesian model that includes all nested haplogroups and accounts for non-independence by modelling their genealogical relationship.

As results from WP3 (chromatin analysis) become available, we will associate Y-haplotype-specific GWAS hits with genes at A/B compartment borders that are only accessible in either X- or Y-bearing spermatids to directly test whether the chromatin context explains haplogroup-dependent associations. Using hitchhiking estimates from WP2, we will estimate the population-averaged polygenic risk score explained by drive-elevated frequencies of risk variants, quantifying the disease burden attributable to meiotic drive. Exploiting our access to UK Biobanks 500,000 fully sequenced participants, each with an ASD spectrum score, we will replicate our analyses to quantify how sex chromosome conflicts influence neurodevelopment across a general population.

Finally, integrating results across all four work packages, we will use the deep learning method (developed in WP4) to predict a male's polygenic risk score from autosomal and X-linked variants conditional on his Y chromosome repeat profile, thus moving from haplogroup categories to a continuous structural predictor.



**Work Package 2: Detection and characterization of meiotic drive.**

While WP1 tests statistical predictions of the drive hypothesis in clinical data, WP2 seeks direct evidence of the drive mechanism and its evolutionary history. We will run genealogy-based selection scans on the X chromosome using tools that reconstruct the genealogical history of individual genomic regions from population samples. Exploiting my extensive expertise in this area, we will use both existing and newly devised genealogy statistics to identify genes subject to strong positive selection and to estimate the extent of the hitchhiking genomic neighborhood.

Using existing single-cell RNA sequencing data across spermatogenesis stages, we will cluster all X-linked genes by expression trajectory to isolate genes not suppressed by meiotic sex chromosome inactivation and genes re-expressed after the meiotic division. Among these genes, we will identify candidate driver genes whose encoded products are not evenly distributed between X and Y-carrying spermatids – despite the active transport across cytoplasmic bridges connecting spermatids meant to ensure this.

For each ASD-associated variant (from WP1), we will estimate the change in allele frequency attributable to hitchhiking, and compute the GWAS effect size stratified by population frequency to estimate how much hitchhiking elevates the population-averaged polygenic risk score — directly quantifying the disease burden that drive imposes beyond what mutation-selection balance alone would produce.

The three-way arms race of Hypothesis 1 applies to primates in general. Exploiting my expertise in comparative genomics we will repeat the selection scans in 500 baboon genomes spanning six species of baboons. Each species represents a natural experiment presenting an independent realization of the process studied in humans. Identifying drivers shared by humans and baboons will greatly bolster our confidence in individual candidate genes. Finding drivers unique to humans or baboons will support the proposed evolutionary mechanisms if such genes meet the expectations of Hypothesis 1.

**Work Package 3: 3D chromatin architecture in spermatogenesis.** This package provides the physical-chemistry dimension of the project, connecting the statistical genetic signals from WP1 and the evolutionary signals from WP2 to the three-dimensional organization of chromosomes in the nucleus.

We will reanalyze published single-cell Hi-C data separating X and Y carrying human sperm to map A/B compartment boundaries across the

genome using Bayesian changepoint detection. This method will provide posterior probability distributions over positions of each boundary and allow us to statistically evaluate if the positions differ in X and Y carrying sperm. We will reanalyze published Hi-C data from neuron organoids to evaluate overlap of A/B borders to those in sperm and to identify neuron specific variable borders. We will characterize all boundaries with respect to gene content, gene expression, CTCF motifs, insulator protein binding, and chromatin super-looping to build a mechanistic model of how a three-way arms race for chromatin control may explain variable borders.

Using a recently published single-cell ATAC-seq atlas of the human brain ([Kim et al. 2024](#)), which profiles chromatin accessibility across cell types in individuals with known Y haplogroup, we will test whether Y haplogroup also influences X-linked and autosomal chromatin structure in neural tissue. If our hypothesis is correct, males should show systematically altered accessibility at the A/B compartment boundaries variable in sperm. This would link the spermatogenesis-focused mechanistic model to a clinical phenotype and could help explain the special role the X chromosome seems to play in ASD-relevant brain structure.

**Work Package 4: Y chromosome repeat content and prediction.** While WP1 uses categorical haplogroup assignments as a proxy for Y structure, this package directly characterizes the continuous variation in Y chromosome repeat content across the 500,000 fully sequenced individuals in UK Biobank and develops machine learning methods to exploit it as a predictive resource. We will profile the repeat content of each Y chromosome by counting instances of all DNA sequence segments of length K (Kmers). Such k-mer profiling has successfully predicted autosomal chromatin state in fruit flies ([Lemos, Araripe, and Hartl 2008](#)).

We will build a fine-scale Y-chromosome haplotype tree using non-repeat sequence (functional single-copy genes) and use a Bayesian regression to model repeat profiles as Brownian motion along its branches. This way we will be able to estimate whether natural selection imposes limits to Y chromosome repeat content and, if so, what characterizes these limits. Britain has a large Indian population, with a Y-haplogroup composition that differs from the native British. We will use standard population genomic methods to assign individual genomic regions to Indian or British ancestry and test the expectation that A/B boundaries on the X chromosome are more strongly affected when the ancestry of the X chromosome region does not match that of the Y chromosome - suggesting that British and Indian populations maintained even transmission through different allele frequency equilibria now disrupted upon admixture.

Third, we will develop a deep convolutional neural network (CNN) that takes a Y chromosome's repeat and ampliconic gene profile as input and predicts chromatin compartment status (A or B) at the compartment borders identified in WP3. Standard linear models cannot capture the complex, nonlinear relationship between the high-dimensional repeat landscape of the Y and chromatin state elsewhere in the genome. A CNN trained on paired Y-structure and chromatin-state data can learn these relationships end-to-end. Successful prediction would provide the strongest evidence that Y structural variation causally influences genome-wide chromatin organization, and would enable individual-level prediction of which genomic regions are epigenetically exposed — the key input to the Y-conditional risk scoring developed in WP1.

**Feasibility and risk management.** I will devote the majority of research time to this project throughout the grant period. The project builds on established data access: the Danish clinical cohort is available through an existing collaboration agreement with a psychiatric genetics group that provides genotyping, quality control, and data governance, while the principal investigator's group contributes the evolutionary genomic and computational expertise. UK Biobank access is secured under an approved application. All computationally intensive analyses will run on the high-performance computing cluster GenomeDK located at Aarhus University. Although extensive interactions and exchange of results between work packages, make the project plan more easily perturbed by negative results, its collaborative nature also provides robustness in terms of publications by individual postdocs/PhD students. However, with the pilot results described, it seems highly likely that the core goals will be met.

## Creativity, Ambition, and Originality

Three aspects of this project are genuinely new. First, it proposes that intragenomic conflict — a fundamental evolutionary force studied primarily in model organisms — has direct, quantifiable consequences for a specific human phenotype. Meiotic drive has never been connected to human disease in this way. The prediction that selfish sex-linked genes continuously generate ASD-relevant variation represents a new explanatory framework, distinct from the standard model in which disease variants are maintained by mutation-selection balance alone.

Second, it introduces the Y chromosome as an active modulator of autosomal gene expression and ASD risk through chromatin regulation. The Y has been entirely absent from ASD genetics. If haplogroup-dependent epistasis proves real, it would provide a mechanistic basis for

the male bias of ASD beyond simple X-linkage, and would explain a measurable fraction of ASD's "missing heritability" — the gap between estimated total heritability and what current GWAS variants explain.

Third, the project bridges evolutionary population genetics, comparative primate genomics, 3D chromatin biology, deep learning, and psychiatric cohort analysis — fields that rarely interact. The replication of selection scans across humans and other primates tests the generality of the conflict, while clinical cohort work tests its medical significance. The methods developed here — X-chromosome GWAS frameworks, Y-stratified heritability models, genealogical statistics for transient drive, and CNN-based chromatin prediction from Y profiles — will be applicable to any complex trait.

## Future Applications in Life Sciences / Health Sciences

Recognizing that evolutionary conflict on the sex chromosomes shapes ASD risk could identify new risk genes and pathways invisible to current approaches. Y-haplotype-stratified polygenic risk scores may improve ASD risk prediction in males — one of the first applications of sex chromosome structural variation in clinical genomics. The CNN-based prediction framework, which maps Y chromosome structure to chromatin state and polygenic risk score, could enable stratification of ASD into subtypes with distinct etiologies — a prerequisite for any future precision-medicine approach. The X chromosome GWAS methods developed here will be broadly useful for any GWAS — most current studies claiming "genome-wide" coverage exclude the X entirely. More broadly, the project establishes a framework for understanding how intragenomic conflict contributes to human disease, with potential relevance to other sex-biased conditions including attention deficit hyperactivity disorder, schizophrenia, and autoimmune disorders.

## Abbreviations

ASD: Autism Spectrum Disorder; GWAS: Genome-Wide Association Study; MSI: Meiotic Sex Chromosome Inactivation (silencing of sex chromosomes during sperm cell production); Hi-C: High-throughput Chromosome Conformation Capture (method for mapping 3D chromosome structure); CNN: Convolutional Neural Network

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## 4. LAY PROJECT DESCRIPTION (max 1,000

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In sexually reproducing species, genes on the sex chromosomes can cheat: a gene on the X that helps its sperm cell outcompete sperm carrying the Y spreads even if it harms the organism. Mathematical models predict that this “meiotic drive” triggers a continuous arms race. The Y and other chromosomes evolve countermeasures, which themselves carry side effects. In this arms race, one of the Y chromosome’s weapons is its bulk of repetitive DNA, which can redirect proteins controlling which genes are switched on — allowing the Y to fight back by shifting gene activity across the genome, including in the brain. This project proposes that this arms race is a major overlooked source of genetic variation underlying autism, which is four times more common in boys than girls. Combining evolutionary modeling, 3D chromosome analysis, deep learning, and data from more than 22,000 autism cases, we test whether sex chromosome conflict shapes the genetic basis of a major neurodevelopmental condition.

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## 5. Anonymized REFERENCES (to be mapped)

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## NOTES FOR DISCUSSION

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1. **v6 changes from v5:** Research Plan rewritten to match restructured 4-WP plan from WP\_detailed\_plans.md. Key changes: (a) WP1 now explicitly follows the 8-item plan with Bayesian haplogroup model, genetic correlation, and cross-WP links to A/B compartments and hitchhiking PRS; (b) WP2 adds SINGER alongside Relate, and includes explicit computation of PRS inflation from hitchhiking; (c) WP3 specifies single-cell Hi-C reanalysis and adds scATAC-seq brain atlas reference ([Kim et al. 2024](#)); (d) WP4 completely rewritten — now focused on Y chromosome ampliconic gene profiling, selection on repeat content, and deep CNN for TAD chromatin prediction from Y profiles (replaces old evolutionary modeling + clinical prediction package). Formal population genetic modeling is now distributed across WP1 (Bayesian heritability, PRS methods) and WP2 (genealogical statistics, hitchhiking quantification). Added CNN, TAD, scATAC-seq, XWAS to abbreviations.

## References

- Brown, Emily J., Alison H. Nguyen, and Doris Bachtrog. 2020. "The Drosophila Y Chromosome Affects Heterochromatin Integrity Genome-Wide." *Molecular Biology and Evolution* 37 (10): 2808–24.  
<https://doi.org/10.1093/molbev/msaa082>.
- Fisher, Ronald A. 1930. *The Genetical Theory of Natural Selection*. Oxford: Clarendon Press.
- Garner, Joseph P., and John P. Capitanio. 2023. "Rhesus Macaque Social Functioning Is Paternally, but Not Maternally, Inherited by Sons: Potential Implications for Autism." *Molecular Autism* 14: 44.  
<https://doi.org/10.1186/s13229-023-00573-2>.
- Kim, Samuel S., Buu Truong, Karthik Jagadeesh, Kushal K. Dey, Amber Z. Shen, Soumya Raychaudhuri, Manolis Kellis, and Alkes L. Price. 2024. "Leveraging Single-Cell ATAC-seq and RNA-seq to Identify Disease-Critical Fetal and Adult Brain Cell Types." *Nature Communications* 15: 563.  
<https://doi.org/10.1038/s41467-024-44742-0>.
- Lemos, Bernardo, Luciana O. Araripe, and Daniel L. Hartl. 2008. "Polymorphic Y Chromosomes Harbor Cryptic Variation with Manifold Functional Consequences." *Science* 319 (5859): 91–93.  
<https://doi.org/10.1126/science.1148861>.
- Lindholm, Anna K., Kelly A. Dyer, Renée C. Firman, Lila Fishman, Wolfgang Forstmeier, Luke Holman, Hanna Johannesson, et al. 2016. "The Ecology and Evolutionary Dynamics of Meiotic Drive." *Trends in Ecology & Evolution* 31 (4): 315–26. <https://doi.org/10.1016/j.tree.2016.02.001>.
- Matos, Bárbara, Stephen J. Publicover, L. Filipe C. Castro, Pedro J. Esteves, and Margarida Fardilha. 2021. "Brain and Testis: More Alike Than Previously Thought?" *Open Biology* 11 (6): 200322. <https://doi.org/10.1098/rsob.200322>.
- Rhie, Arang, Sergey Nurk, Monika Cechova, Savannah J. Hoyt, Dylan J. Taylor, Nicolas Altemose, et al. 2023. "The Complete Sequence of a Human Y Chromosome." *Nature* 621: 344–54. <https://doi.org/10.1038/s41586-023->



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